Aaditya Vikram Saravana Bhavan

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Computational Data Science (MCDS), GPA: 4.00/4.00 *Dec 2026*
Relevant Courses: Introduction to Deep Learning (PhD), Introduction to Machine Learning, Foundations of Computational Data Science, Cloud Computing, Multimodal Machine Learning

PES University Bangalore, India
B.Tech in Computer Science and Engineering, GPA: 9.42/10.00 *May 2024*

Experience

Amazon Web Services Seattle, WA
Software Development Engineer Intern, Amazon RDS *May 2026 – Aug 2026*
- Engineered an idempotent Java RDS control-plane workflow that propagates RSA-2048, RSA-4096, and ECC-384 regional CA certificates across 3 upcoming AWS Regions, enabling TLS-authenticated connections to RDS instances
- Built 2 AWS Lambda functions and extended AWS CDK and IAM infrastructure to move certificate artifacts from a tooling-account S3 bucket to a global publication account, cutting trust-store publication from weeks to days
- Automated internal trust-store onboarding across 2 repositories, 4 branches, and 12 configuration files by retrieving regional certificates, generating code reviews, and integrating failure ticketing into region-build automation

Cisco Chennai, India
Software Development Engineer, Catalyst Center *Sep 2024 – Jul 2025*
- Eliminated 24-hour telemetry gaps across 10K+ customer deployments by rebuilding a Go polling service with per-device checkpoints that backfill missed data after any backup, restore, or outage; validated in AWS Athena
- Built Node.js Webex bots delivering real-time threshold and system-health alerts to on-call engineers, reducing mean time to resolution by approximately 40%
- Led error-contract standardization across 8 engineering teams, replacing inconsistent microservice responses with a shared schema and refactoring backend parsing to surface clear, actionable errors to customers

Zscaler Bangalore, India
Software Development Engineer Intern *Feb 2024 – Aug 2024*
- Reduced data size by 40% and transfer latency by 50% using Protocol Buffers; tuned PostgreSQL and PL/pgSQL
- Secured RabbitMQ communication, automated server workflows with Python, and achieved 95% unit test coverage
- Improved system-level C serialization pipelines and reduced code duplication using X-Macros

Academic Projects

High-Throughput Twitter Recommendation Service Pittsburgh, PA
Carnegie Mellon University, Cloud Computing *Jan 2026 – Apr 2026*
- Built a Twitter recommendation service over 1 TB+ of data using PySpark, Go, Amazon EKS, Kubernetes, and Aurora MySQL, achieving 6,000 peak RPS and 5,000 sustained RPS at 99.7% correctness
- Increased throughput from 200 to 5,000 sustained RPS through schema denormalization, indexed Aurora lookups, horizontal pod scaling, connection-pool tuning, and concurrent authentication and database queries

MyTorch Pittsburgh, PA
Carnegie Mellon University, Intro to Deep Learning *Aug 2025 – Dec 2025*
- Built a PyTorch-style deep learning library from scratch with reverse-mode autograd over a modular computation graph, verifying analytical gradients against numerical references to within 1e-5
- Implemented and trained MLPs, CNNs, RNNs (GRU/LSTM), and Transformers on it, with optimizers and losses

GAN-Based Multispectral Image Dehazing Bangalore, India
PES University *Jan 2023 – Apr 2024*
- Found that state-of-the-art dehazing models fail on multispectral imagery by assuming haze is uniform across wavelengths; built a GAN in TensorFlow that processes each band separately, beating prior methods on SHIA
- Awarded **Best Paper** at ISMSI 2024; improves NDVI accuracy for crop-health assessment in agricultural remote sensing

Distributed Messaging and Stream Processing Bangalore, India
PES University *Nov 2022 – Jan 2023*
- Built a Kafka-like messaging system over raw sockets with a producer–consumer architecture, topic management, and multi-broker replication using ZooKeeper for leader election and failover recovery
- Extended it into stream processing: a Java Kafka producer routing a ride-hailing trace into 2 topics partitioned 5 ways by block ID, feeding stateful Samza jobs that hold per-block driver state and match riders by a four-factor weighted score

Publications

- **Enhancing Multispectral Vision: A GAN-Based Dehazing Framework**, ISMSI 2024
- **A Pugnacious Comparative Study of Data Analysis Techniques for Wine Quality**, I2CT 2024
- **Android Malware Detection: A Comprehensive Review**, Research Advances in Network Technologies 2024

Skills

Languages: Go, C/C++, Python, Java, SQL; **LeetCode Knight (Top 2.75%)**
Cloud & Systems: AWS, Distributed Systems, Microservices, REST APIs, Linux, Docker, Kubernetes, AWS CDK
Databases & Data: Aurora MySQL, PostgreSQL, MongoDB, PySpark, AWS Athena
Machine Learning: PyTorch, TensorFlow, scikit-learn, Transformers, Computer Vision